

# Mining AI Analyst

An AI-powered research assistant for junior mining stock analysis

## The Problem

Junior mining and exploration companies publish dense, promotional investor decks full of technical jargon (NPV, IRR, PEA vs. PFS, resource categories) and selective framing (best-case price decks, partial-deposit valuations). Novice investors have no easy way to verify claims, compare to peers, or spot red flags without significant domain knowledge.

## The Solution

A conversational AI assistant, purpose-built for mining sector analysis, that ingests a company's investor materials and filings, extracts and verifies key metrics, checks claims against the underlying numbers, benchmarks against named peers, and explains everything in plain language for a non-expert investor.

## Core Features

- **Document ingestion** — Upload or auto-fetch investor decks, NI 43-101 technical reports, and SEDAR+/EDGAR filings.
- **Metric extraction** — Automatically pulls NPV, IRR, capex/opex, resource category (Measured/Indicated/Inferred), study stage (PEA/PFS/FS), mine life, ownership & dilution.
- **Claim verification** — Cross-checks promotional statements (e.g. "0.1x NAV") against the underlying figures in the same document.
- **Red-flag detection** — Flags common patterns: stale studies, partial-deposit valuations, cherry-picked price decks, undisclosed dilution risk.
- **Peer comparison** — Benchmarks the company against a curated database of comparable projects (same commodity, jurisdiction, or stage).
- **Plain-language education** — Explains mining/financial terminology and concepts as they come up, tailored to a beginner.

## Technical Approach

| Component               | Role                                                                                     |
|-------------------------|------------------------------------------------------------------------------------------|
| System prompt / "Skill" | Encodes the analysis workflow: fetch → extract → verify → compare → flag risks → explain |
| RAG knowledge base      | Mining glossary, NI 43-101 standards, historical peer comps, red-flag pattern library.   |
| Tool integrations       | SEDAR+/EDGAR fetchers, live commodity price data, document parsing.                      |
| Claude API              | Powers reasoning, extraction, and conversational explanation.                            |

## Why Focus on Mining

- Standardized reporting (NI 43-101, PEA/PFS/FS stages) makes consistent automated analysis feasible.

- Highly promotional investor decks create real demand for independent claim-checking.
- Passionate, underserved retail audience (forums, YouTube, junior-resource communities).
- No existing AI tool specializes in this niche — competitors are generalist (ChatGPT, Gemini) or quant/signal-based (Danelfin, Kavout), not mining-specific or document-verification focused.

## **Suggested MVP Scope**

Start narrow: one commodity (e.g. iron ore or gold juniors), one workflow (deck/filing upload → structured analysis report), and a small but high-quality peer-comps database. Expand commodities and data sources after validating the core verification workflow with real users.